

GUILLAUME GENOIS

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EDUCATION

Master's Degree in Machine Learning

2025 – Expected Dec. 2026

Mila - Quebec Artificial Intelligence Institute

Montréal, QC

Bachelor's Degree in Mathematics and Computer Science

2022 – 2025

Université de Montréal

Montréal, QC

Awarded excellence scholarships and research funding

PROFESSIONAL EXPERIENCE

Data Science Intern

May 2026 – Dec. 2026

Bell Canada

Montréal, QC

- ❖ Built an end-to-end anomaly detection project monitoring QoE KPIs with automatic notifications
- ❖ Estimated to save 10 hours weekly of manual work, yielding \$70k in annual savings on a \$1k budget
- ❖ Designed the training and in-production infrastructure leveraging GCP, BigQuery and Agent Platform
- ❖ Pipeline ingests millions of crowdsourced network records daily (latency, throughput, video buffering)
- ❖ Scientifically evaluated SOTA ML models such as CNNs, autoencoders and iForests on benchmarks
- ❖ Presented the solution to senior managers and executives, securing adoption by the Best Network team

Cloud and AI Solutions Advisor

Jun. 2025 – Apr. 2026

Travel Direct Group

Montréal, QC

- ❖ Led technical review of new AI-powered virtual assistant project for personalized offers
- ❖ Initiated an internal process automation project using self-hosted open-source n8n and an LLM
- ❖ Deployed 80+ websites using Azure cloud infrastructure with Kubernetes, CDN and WordPress
- ❖ Assisted clients with DNS modifications, providing guidance and support for website deployments
- ❖ Identified system bugs and coordinated resolution with external partners for Meta API integrations
- ❖ Represented the company at ALL IN AI, driving the first phase of AI partner discovery

Data Engineering Advisor

May 2024 – Apr. 2025

Desjardins

Montréal, QC

- ❖ Developed a Python parser to extract field usage and sources from SQL and SAS queries
- ❖ Automated data traceability, reducing manual impact analysis time from days to minutes
- ❖ Contributed to multidimensional model development using CI/CD practices
- ❖ Implemented modernization solutions with Azure DevOps and SQL Server integration

System Pilot

May 2023 – Apr. 2024

Québec Office of the French Language, Government of Québec

Montréal, QC

- ❖ Responded to internal and media data-driven requests using SQL extraction and analysis
- ❖ Coordinated internal clients and the IT team to resolve and validate system issues

LEADERSHIP & COMMUNITY

Badminton Coach (Beginner to Competitive)

2020 – 2023

Badminton Lachine Club

Lachine, QC

- ❖ Led groups of 20 players, coordinating a team of assistant coaches to plan and run training sessions
- ❖ Set long-term team and individual development goals, adapting instruction to each player's level

Website Lead (Volunteer)

2023 – Present

Club des personnes handicapées du Lac St-Louis

Lachine, QC

- ❖ Took ownership of the organization's legacy website, modernizing it to meet management's needs
- ❖ Update content monthly and lead the migration to a better-suited platform

Volunteer for Harvest and Other Activities

2015 – Present

Vignoble Aux Volets Noirs

Très-Saint-Sacrement, QC

Volunteer for Carlisopen Golf Tournament supporting Fondation En Coeur

2023 – 2024

Lac-St-Joseph Golf Club

Québec, QC

PROJECTS

RAG-Powered Cruise Booking Assistant | *LangChain, Groq, ChromaDB, Hugging Face*

- ❖ Built hybrid RAG system combining SQL database queries with vector similarity search
- ❖ Implemented ReAct agent to orchestrate multi-tool workflows with conversation memory
- ❖ Deployed production web application with Gradio interface and nginx on rag.guillaume.genois.ca

Metadata Curation to Evaluate the Use of LLMs in SLRs | *RPA, HTML Parsing, Dataset*

- ❖ Built automated application to extract over 30,000 scientific articles from 8 major academic databases
- ❖ Developed custom HTML parser to systematically extract metadata from research publications
- ❖ Designed 16 usable datasets for LLM evaluation research on systematic literature review acceleration

Smart Grid Demand Response Optimization | *JAX, Model Predictive Control, Time-Series Forecasting*

- ❖ Modeled building thermal dynamics using RC networks with Kalman filtering on Hydro-Québec data
- ❖ Implemented MPC-based heating optimization to minimize electricity costs while maintaining comfort
- ❖ Achieved a 19.3% peak demand reduction through staggered pricing, flattening rebound effects

SKILLS

Languages: French (native), English (fluent)

Machine Learning: Python, RL, MPC, RAG, Agentix, PyTorch, TensorFlow, JAX, Scikit-learn

Data & Software Engineering: Excel, SQL, SAS, Matlab, Azure, Docker, Kubernetes, CI/CD, Agile, Jira

Strategy & Advisory: Business-AI Translation, Executive Communication, Governance, ROI Evaluation

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